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EXECUTIVE SUMMARY

1.1 Landslide risk assessment

The landslide risk assessment was undertaken for the sub-catchments of Rwandex-Magerwa, Bishenyi, Rwabayanga and Rusizi. The deliverable has been landslide risk maps for each watershed (refer to pdf maps accompanying report). Maps are classified in several classes.

The landslide risk maps are provided in the form:

- Landslide inventories (included in accompanying PDF maps);
- Landslide susceptibility maps (included in accompanying PDF maps);
- Landslide hazard maps (included in accompanying PDF maps);
- Exposure databases (included in report)

Susceptibility Class	Bishenyi		Rwabayanga		Rusizi		Rwandex-Magerwa	
	Area (km^2)	Percentage of total area	Area (km^2)	Percentage of total area	Area (km^2)	Percentage of total area	Area (km^2)	Percentage of total area
<= 0.2	9.87	20.9%	2.54	32.5%	9.50	45.1%	4.45	45.8%
0.2 - 0.45	14.85	31.4%	3.06	39.1%	6.55	31.1%	2.83	29.1%
0.45 - 0.55	12.47	26.4%	1.20	15.3%	3.31	15.7%	1.29	13.3%
0.55 - 0.8	7.45	15.8%	0.67	8.6%	1.45	6.9%	0.73	7.5%
0.8 - 1.0	2.62	5.5%	0.35	4.5%	0.25	1.2%	0.41	4.2%

Results for landslide susceptibility are summarised in the table above. The continuous values of the susceptibility models are classified into five unequally-spaced susceptibility classes. The category [0.80 – 1.0] presents the class that is the most prone to landsliding. The opposite class is ≤ 0.2]. The class]0.45-0.55] present the zone where the uncertainty on the model classification performance is the highest (Rossi et al., 2010).

Results for landslide hazard are summarised in the table below.

Hazard rate class (m^2/year/km^2)	Bishenyi		Rwabayanga		Rusizi		Rwandex-Magerwa	
	Area (km^2)	Percentage of total area	Area (km^2)	Percentage of total area	Area (km^2)	Percentage of total area	Area (km^2)	Percentage of total area
1	21.06	44.6%	5.34	68.3%	15.98	75.9%	6.77	69.7%
18	19.54	41.3%	1.84	23.5%	4.40	20.9%	2.19	22.5%
90	5.96	12.6%	0.59	7.5%	0.66	3.1%	0.68	7.0%
365	0.71	1.5%	0.05	0.6%	0.02	0.1%	0.08	0.8%